

Solving Quadratic Equations by Extracting Square Roots

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A **quadratic equation** can sometimes be solved
by isolating the squared expression first.

Square Root Property

If an equation can be written in the form

$$x^2 = a,$$

then apply the **square root property**:

$$x = \pm\sqrt{a}.$$

Perfect Square Expression

If the equation has the form

$$(x - h)^2 = a,$$

then

$$x - h = \pm\sqrt{a},$$

and solve for x .

Two Real Solutions

If $a > 0$, the equation has **two real solutions**.

$$x = \pm\sqrt{a}$$

The two solutions are

$$x = \sqrt{a}$$

and

$$x = -\sqrt{a}.$$

One Real Solution

If $a = 0$, the equation has **one real solution**.

$$x = \pm\sqrt{0}$$

Therefore,

$$x = 0.$$

No Real Solutions

If $a < 0$, the equation has **no real solutions**.

$$x = \pm\sqrt{a}$$

The square root of a negative number is not a real number.

Checking Solutions

Always **check your solutions** by substituting them into the original equation.

This confirms that the solutions satisfy the given equation.

Examples

Example 1

Solve

$$x^2 = 49.$$

Take the square root of both sides:

$$x = \pm\sqrt{49}$$

$$x = \pm 7.$$

Therefore,

$$x = 7 \quad \text{or} \quad x = -7.$$

Example 2

Solve

$$(x - 4)^2 = 25.$$

Take the square root of both sides:

$$x - 4 = \pm\sqrt{25}$$

$$x - 4 = \pm 5.$$

Therefore,

$$x = 9 \quad \text{or} \quad x = -1.$$

Example 3

Solve

$$2(x - 1)^2 = 98.$$

Divide by 2:

$$(x - 1)^2 = 49.$$

Take the square root:

$$x - 1 = \pm 7.$$

Therefore,

$$x = 8 \quad \text{or} \quad x = -6.$$

Practice Exercises

Practice 1

Solve the quadratic equation.

$$x^2 = 49$$

Practice 2

Solve the quadratic equation.

$$x^2 = 144$$

Practice 3

Solve the quadratic equation.

$$x^2 = \frac{81}{16}$$

Practice 4

Solve the quadratic equation.

$$(x - 4)^2 = 25$$

Practice 5

Solve the quadratic equation.

$$(x + 3)^2 = 64$$

Practice 6

Solve the quadratic equation.

$$3x^2 = 108$$

Practice 7

Solve the quadratic equation.

$$2(x - 1)^2 = 98$$

Practice 8

Solve the quadratic equation.

$$(2x + 1)^2 = 49$$

Practice 9

Solve the quadratic equation.

$$(3x - 6)^2 = 144$$

Activities

Activity 1

Solve the quadratic equation.

$$x^2 = 64$$

Activity 2

Solve the quadratic equation.

$$x^2 = 225$$

Activity 3

Solve the quadratic equation.

$$x^2 = \frac{121}{25}$$

Activity 4

Solve the quadratic equation.

$$(x - 6)^2 = 16$$

Activity 5

Solve the quadratic equation.

$$(x + 5)^2 = 81$$

Activity 6

Solve the quadratic equation.

$$5x^2 = 320$$

Activity 7

Solve the quadratic equation.

$$4(x + 2)^2 = 144$$

Activity 8

Solve the quadratic equation.

$$(2x - 3)^2 = 81$$

Activity 9

Solve the quadratic equation.

$$(4x + 8)^2 = 256$$

Thank you!